1. Mathematics Assignment Solutions

Problem 1: The Jacobian Matrix

Given $y_1 = x_1^2 x_2$ and $y_2 = 5x_1 + \sin(x_2)$.

Task A (First-order partial derivatives):

$$\frac{\partial y_1}{\partial x_1} = 2x_1 x_2 \qquad \frac{\partial y_1}{\partial x_2} = x_1^2$$

$$\frac{\partial y_2}{\partial x_1} = 5 \qquad \frac{\partial y_1}{\partial x_2} = \cos(x_2)$$

Task B (Construct $J(x)$):

$$J(x) = \begin{bmatrix} 2x_1 x_2 & x_1^2 \\ 5 & \cos(x_2) \end{bmatrix}$$

Task C (Evaluate at $x_1 = 2, x_2 = 0$):

$$J(2,0) = \begin{bmatrix} 2(2)(0) & (2)^2 \\ 5 & \cos(0) \end{bmatrix} = \begin{bmatrix} 0 & 4 \\ 5 & 1 \end{bmatrix}$$

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2

Problem2: The Hessian Matrix

Given $L(w_1, w_2)\ w_1^3 + 2w_1 w_2^2 - 4w_2$.

Task A (The Gradient):

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix} = \begin{bmatrix} 3w_1^2 + 2w_2^2 \\ 4w_1 w_2 - 4 \end{bmatrix}$$

Task B (Second Derivatives):

$$\frac{\partial^2 L}{\partial w_1^2} = \frac{\partial}{\partial w_1}(3w_1^2 + 2w_2^2)\ 6w_1$$

$$\frac{\partial^2 L}{\partial w_1 \partial w_2} = \frac{\partial}{\partial w_2}(3w_1^2 + 2w_2^2)\ 4w_2$$

$$\frac{\partial^2 L}{\partial w_2 \partial w_1} = \frac{\partial}{\partial w_1}(4w_1 w_2 - 4)\ 4w_2$$

$$\frac{\partial^2 L}{\partial w_2^2} = \frac{\partial}{\partial w_2}(4w_1 w_2 - 4) = 4w_1$$

Task C (Construct H): The mixed partials match ($4w_2 = 4w_2$). The matrix is perfectly symmetric

$$H = \begin{bmatrix} 6w_1 & 4w_2 \\ 4w_2 & 4w_1 \end{bmatrix}$$

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Problem 3: Curvature Classification.

Task A (Evaluate H at $w_1 = 1, w_2 = 0$):

$$H(1,0) = \begin{bmatrix} 6(1) & 4(0) \\ 4(0) & 4(1) \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix}$$

Task B (Eigenvalues): Since the matrix is diagonal the eigenvalues are simply the entries on the main diagonal: $\lambda_1 = 6$, $\lambda_2 = 4$.

Task C (Classification): Since both eigenvalues are strictly Positive (>0), the matrix is Positive Definite. Therefore, this Critical Point is a Local Minimum (a bowl shape)